BAYU SETIAWAN

SOFTWARE ENGINEER & AI ENGINEER

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PROFILE SUMMARY

Fresh Graduate Software Engineer and AI Engineer, and a recent Computer Science graduate (GPA 3.31) with a strong track record in building end-to-end, data-driven applications. Experienced in developing Machine Learning systems for the manufacturing industry, including a production forecasting model (Random Forest, R^2 0.81) that optimized planning across 10 product categories, managing the full lifecycle from data preprocessing to deployment via Flask, Laravel, and Flutter. Recently expanded expertise into Full-Stack AI development by architecting an LLM-powered SaaS platform (Next.js, FastAPI) featuring advanced prompt engineering, real-time unstructured data parsing, and scalable cloud deployment (Vercel, Railway). Highly comfortable bridging the gap between complex AI models and production-ready web and mobile applications. Eager to contribute as a Software Engineer or AI Engineer at a product-oriented, data-driven company.AI Engineer.

WORK EXPERIENCE

Software Engineer Intern | PT Honoris Industry *August 2025*

Vehicle Fleet Cubication Selection System (Logistics Optimization)

- Designed a mobile-based fleet cubication system using Dart (Flutter) to help operational teams select optimal vehicles for each shipment, reducing unused cargo space and improving fleet selection accuracy.
- Implemented a real-time volume calculation algorithm within the Flutter application to give field operations instant fleet recommendations on-site, cutting manual calculation time and streamlining vehicle selection during shipment.

Machine Learning-Based Production Volume Prediction System

- Built an end-to-end machine learning system to forecast monthly production needs for 10 LED product categories, achieving an R^2 of 0.814 and low MAPE on 86 months of historical test data (Jan 2019 - Feb 2026), using Random Forest Regression.
- Designed a data preprocessing pipeline and tested 4 feature engineering approaches to identify the most business-realistic prediction model, validated with TimeSeriesSplit cross-validation for reliable performance estimation.
- Developed an interactive dashboard to give management real-time visibility into production forecasts and reduce overproduction risk by an average of ~3,900 units/month, built with Flask and integrated automated production-control logic.
- Collaborated with academic advisors and internal company stakeholders to align the model with real operational requirements, ensuring the system was production-ready before deployment.

HI Inventory System Redesign (Web to Mobile Migration)

- Redesigned PT Honoris Industry's HI Inventory system from a web-based platform into a mobile application to improve accessibility and usability for field use, built using Flutter, Laravel, and MySQL.
- Developed a RESTful API to connect the Flutter application with the Laravel backend, enabling real-time data synchronization and centralized inventory data management through MySQL.

KEY PROJECTS

LamarKerja.AI (AI-Powered CV Optimizer & Cover Letter Generator) *September 2026 – October 2026*

- End-to-End SaaS Development: Architected and developed a full-stack platform enabling users to upload CVs, analyze ATS compatibility scores, and generate tailored cover letters in real-time.
- AI Integration & Prompt Engineering: Integrated Groq Cloud API (Llama 3) for CV data extraction and text generation. Engineered robust prompt strategies with fallback and auto-retry mechanisms, reducing AI JSON parsing errors to 0%.
- Unstructured Data Processing: Built an asynchronous backend pipeline using PyMuPDF to accurately extract text from PDF files and map it to structured JSON for seamless PostgreSQL storage via SQLAlchemy.
- UX/UI Optimization: Implemented a mobile-first, responsive design using Tailwind CSS, ensuring flawless adaptation across all devices with dark mode support and native-like navigation.
- DevOps & Security: Deployed a decoupled architecture to production via Vercel (Frontend) and Railway (Backend), securing API endpoints with JWT authentication and strict CORS configurations.
- Tech Stack: Next.js, React, TypeScript, Tailwind CSS, FastAPI, Python, PostgreSQL (Neon), Groq API (Llama 3), PyMuPDF, Zustand.

ADDITIONAL PROJECTS

Freelance *2023 - 2026*

- Developed a CRUD-based food ordering website using Laravel to manage menu, orders, and inventory data efficiently.
- Built a personal portfolio website for a client using native PHP to showcase their work and professional profile.
- Redesigned the Bapenda Kota Bogor website using native PHP to improve structure and usability.
- Built a personal portfolio website using React (JSX), Tailwind CSS, and HTML/CSS to showcase professional skills and projects with a responsive design.

EDUCATION

Bachelor of Computer Science | Universitas Pakuan *2022 - 2026*
GPA: 3.31/4.00.
Thesis: "Implementation of the Random Forest Algorithm in a Web-Based Lamp Production Information System for Production Quantity Prediction at PT Honoris Industry."

SKILLS

- Programming Languages: Python, JavaScript, TypeScript, PHP, Dart, HTML5/CSS3
- Frontend Development: Next.js (App Router), React.js, Tailwind CSS, Zustand (State Management), Responsive Design
- Backend Development: FastAPI, Flask, Laravel, RESTful API Design, Asynchronous Programming (async/await), Pydantic
- Database & ORM: PostgreSQL, MySQL, SQLAlchemy (Async ORM), Neon (Serverless DB)
- AI & Data Science: LLM Prompt Engineering, Groq API, Scikit-learn, Pandas, Random Forest Regression, Time Series Cross-Validation, PyMuPDF (PDF Text Extraction)
- DevOps & Tools: Git, GitHub, Vercel, Railway, Postman, Google Colab, VS Code, Chrome DevTools
- Soft Skills: Problem Solving, Analytical Thinking, Teamwork, Technical Communication, Time Management

CERTIFICATIONS

Agentic AI Executive Micro-Certification | ServiceNow *October 2026*

- Completed comprehensive micro-certification focusing on Agentic AI concepts, executive-level AI integration, and leveraging AI to transform modern workflows.

Agentic AI 'Smart Student with AI' Certification | ComSTraC (Computing Centre), Universitas Pakuan

- Certified in Agentic AI fundamentals, focusing on enhancing student competencies and productivity through AI tools in the digital era.

Committee Member – Seminar 2024: | Department of Education, BEM FMIPA, Universitas Pakuan *October 2024*

- Organized and managed a university-wide seminar, collaborating with faculty and student leaders to provide educational guidance to peers, demonstrating strong teamwork and event management skills.